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TOBY STUART

DAVID KIRON

Sirtris Pharmaceuticals: Living Healthier, Longer

"You can live to be a hundred if you give up all the things that make you want to live to be a hundred."
—Woody Allen

One Saturday in February 2007, Dr. David Sinclair and Dr. Christoph Westphal co-founders of Sirtris Pharmaceuticals, a Cambridge, MA-based life sciences firm, navigated the company's narrow hallways and cramped offices to a conference room for their regular weekend strategy planning session.

When they reached the conference room, Sinclair and Westphal reviewed their activities during the past week. Sinclair, who was an associate professor of pathology at Harvard Medical School and co-chair of Sirtris's Scientific Advisory Board, had had interviews with Charlie Rose, the *Wall Street Journal,* and *Newsweek*. Westphal, who was Sirtris's CEO and vice chairman, had closed a $39 million round of financing, bringing the total amount of invested capital in the company to $103 million.

Sinclair and Westphal were riding a wave of interest generated, in part, by their company's promising research into age-related diseases, such as diabetes, cancer, and Alzheimer's. The company's research into disease, however, only partly explained its appearance on the covers of *Scientific American, Fortune*, and *the Wall Street Journal*. According to their suggestive headlines – "Can DNA Stop Time: Unlocking the Secrets of Longevity Genes" (*Scientific American*), "Drink wine and live longer: The exclusive story of the biotech startup searching for anti-aging miracle drugs" (*Fortune*) and "Youthful Pursuit: Researchers seek key to Antiaging in Calorie Cutback" (*Wall Street Journal*) – Sirtris was hoping to develop drugs that could treat diseases of aging, and in so doing had the potential to extend the lifespan of human beings.[1]

The Sirtris team had, in fact, established a link between resveratrol, a compound found in red wine-producing grapes, and sirtuins, a newly discovered family of enzymes with links to improved longevity, metabolism and health in living things as diverse as yeast, mice and humans. Sinclair and Westphal were building Sirtris around the development of sirtuin-activating drugs for the diabetes

[1] Leonard Guarente and David Sinclair, "Can DNA Stop Time: Unlocking the Secrets of Longevity Genes," *Scientific American*, March 2006; David Stipp, "Researchers seek key to antiaging in calorie cutback," *Wall Street Journal*, October 30, 2006. David Stipp, "Drink wine and Live Longer: The exclusive story of the biotech startup searching for anti-aging miracle drugs," *Fortune*, February 12, 2007. See **Appendix** for cover of the *Wall Street Journal* article.

Professor Toby Stuart and Senior Researcher David Kiron, Global Research Group, prepared this case, with advice and contributions from Alexander Crisses (MBA 2008). HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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market. The Sirtris team had developed its own proprietary formulation of resveratrol, called SRT501, and was developing new chemical entities (NCEs) that were up to 1000x more potent activators of sirtuins than resveratrol.

In today's strategy session, which included Dr. Michelle Dipp, Sirtris's senior director of corporate development and Garen Bohlin, the company's chief operating officer, the team was discussing their upcoming move to new laboratory space in another part of Cambridge, and three of the more pressing strategic issues facing the firm.

- **In-licensing.** One issue was whether to in-license compounds from a biotech company to diversify Sirtris's drug development platform beyond its narrow focus on SIRT1, one of seven sirtuin variants in the human body. Several members of the Sirtris executive team were advocating a more balanced risk portfolio as the company started to increase investment in its drug development efforts.

- **Partnership with Pharma**. As is almost always the case in biotech, the team was in discussions about a partnership with a few large pharmaceutical firms. They were considering (a) what it would mean for the organization to become tied to a pharmaceutical company at this stage of its development and (b) whether to postpone a deal until Sirtris had more clinical data. Was this the right point in the company's history to do a deal?

- **Nutraceuticals**. Sinclair received a near-constant stream of emails and phone calls from the public requesting Sirtris's proprietary version of resveratrol, SRT501. For some time, he had contemplated selling SRT501 as a nutraceutical, an off-the-shelf health supplement that would not require FDA approval. This idea raised many questions about market opportunity, commercialization strategies, and the potential impact of a nutraceuticals offering on the Sirtris brand and the all-star group of scientists that had allied themselves with the organization.

Anti-Aging Science

The quest for long life has spurred the imagination of people in virtually every era in human history. Ancient Greeks imagined immortal gods, the sixteenth century Spanish adventurer Ponce de Leon searched for the fountain of youth, and twenty-first-century scientists test rodents for life-extending biological compounds.

Until the 1990s, the only proven means of increasing life-span in any animal was to reduce its calorie intake. In 1935, Cornell University researchers discovered that reducing calorie intake in rodents by 40% increased their lifespan by an average of 30-40%. Conventional wisdom became that calorie reduction (CR) activates an evolutionary adaptive process that lowers metabolism to help animals through periods of food shortages or droughts.

Decades passed before scientists could shed light on the biological mechanism triggered by CR. When new information arrived in the 1980s and early 1990s, it contradicted what had become the conventional wisdom. The new work indicated that instead of lowering metabolism, calorie reduction is a biological stressor that activates a defensive metabolic response. Few scientists paid much attention to the shift in view, as few serious scientists focused their academic careers on anti-aging studies.

Longevity research began to gain traction as an academically credible field of study in the early 1990s, after MIT professor Leonard Guarente and his laboratory traced the molecular pathway of calorie reduction in yeast to sirtuins (silent information regulators), which are proteins (enzymes) that

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are found in all cells. (See **Exhibit 1** for timeline of scientific milestones in longevity research.) In humans, there are seven types of sirtuins in different parts of cells and in different parts of the body. Sirtris was focusing 90% of its R&D on one sirtuin, called SIRT1 in humans. For simplicity, any reference to sirtuins, unless otherwise noted, is to the family of sirtuins or SIRT1.

David Sinclair

In 1993 while sightseeing in Sydney, Australia, Guarente was taken to dinner by some local yeast researchers, a group that included David Sinclair, then a young doctoral candidate at the University of New South Wales. During the dinner, Guarente mentioned that he had two students working on aging in yeast. "I was incredibly excited by Lenny's work," said Sinclair. "I asked him why the longevity field was so pre-occupied with looking for genes that ended life, rather than genes that could extend it. By the time we finished dinner, I told him I was going to work with him at MIT."

Sinclair grew up in St. Ives, near Sydney Australia, the eldest son of parents who both worked in the medical diagnostics industry. In high school, he was known as a talented class clown and risk-taker, a young man who aced science classes but could not resist setting off minor explosions in chemistry lab.[2]

Two years after their first meeting, Sinclair made good on his promise and joined Guarente's MIT lab as a postdoctoral fellow. Sinclair quickly established himself as a creative and productive researcher, publishing a 1997 article in *Cell* with Guarente that described how the yeast equivalent of SIRT1 increased the longevity of yeast. When yeast cells divide (a sign of aging in yeast cells), they spin off extra copies of genetic material called extrachromosomal rDNA circles (or ERCs). With each successive cell division, ERC copies accumulate in the nucleus. The original cell, faced with copying both its original genetic material and an increasing number of ERCs, soon runs out of energy and eventually dies. But when an extra copy of the sirtuin gene was added to the cell nucleus, the formation of ERCs was repressed and the cell's life span was extended by 30 percent.

In 1999, Sinclair left Guarente's lab for a tenure track position at Harvard Medical School, but continued to collaborate with Guarente.[3] They found that extra copies of the sirtuin gene extended the life span of roundworms by as much as 50 percent. "We were surprised not only by this commonality in organisms separated by a vast evolutionary distance but by the fact that the adult worm body contains only non-dividing cells," wrote Guarente and Sinclair in their 2006 *Scientific American* article.[4]

The search was on for sirtuin activating compounds, or STACs. This was a high-stakes search. "No chemical or drug had ever increased the activity of sirtruins" said Dr. Dipp, affectionately

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[2] David Stipp, "Drink wine and Live Longer," *Fortune*, February 12, 2007.

[3] Dr. Sinclair obtained a BS with first-class honors at the University of New South Wales, Sydney, and received the Commonwealth Prize for his research. In 1995, he received a Ph.D. in Molecular Genetics and was awarded the Thompson Prize for best thesis work. He worked as a postdoctoral researcher with Dr. Leonard Guarente at M.I.T. being recruited to Harvard Medical School at the age of 29. Dr. Sinclair has received several additional awards including a Helen Hay Whitney Postdoctoral Award, and a Special Fellowship from the Leukemia Society, a Ludwig Scholarship, a Harvard-Armenise Fellowship, an American Association for Aging Research Fellowship, and is currently a New Scholar of the Ellison Medical Foundation. He won the Genzyme Outstanding Achievement in Biomedical Science Award for 2004. http://medapps.med.harvard.edu/agingresearch/pages/faculty.htm, Accessed 1.4.07.

[4] Leonard Guarente and David Sinclair, "Can DNA Stop Time: Unlocking the Secrets of Longevity Genes," *Scientific American*, March 2006.

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known within the firm as "The General". "A compound that could activate sirtuins would increase the speed of cellular metabolism. It could have far reaching implications for human healthcare."

In 2003, Sinclair discovered that resveratrol, a compound found in red wine, activated sirtuins in yeast cells, a discovery which indicated that in fact it might be possible to develop a drug that could activate the sirtuin enzyme. One way to activate the sirtuin enzyme was to optimize the effects of resveratrol by giving it in highly purified form. Another was to mimic the effects of resveratrol using an entirely new, more potent chemical structure. Sirtris was pursuing both approaches through its SRT501 and new chemical entity drug development projects.

When Sinclair's *Nature* article was published September 11, 2003, it was hailed by many scientists as a seminal paper, but it also drew criticism from members of the scientific community, including former colleagues from Guarente's MIT laboratory. The article also drew the attention of Dr. Christoph Westphal, who had recently been promoted to general partner at Polaris Venture Partners, one of the larger Boston-area venture capital funds.

Christoph Westphal

In his four years at Polaris, Westphal had had several successful investments and stints as founding CEO. Between 2000 and 2004, Westphal co-founded five companies and served as the original CEO at four of them. In all cases, Westphal recruited a CEO to replace him and remained on the board as lead investor once he got the company off the starting blocks. Two went public – Alnylam and Momenta – and had a combined market value of $1.4 billion in early 2007. Philip Sharp, a Nobel Prize winning biologist and Sirtris advisor, described Westphal's business and science acumen: "Christoph's combination of skills is very rare. I haven't seen his equivalent in 30 years of working in biotech."[5] In 2002, MIT's *Technology Review* recognized Westphal as one of the country's top 100 Young Innovators under 35.

The son of two doctors, Westphal was a former McKinsey consultant and physician, who sped through an MD/Ph.D. program at the Harvard Medical School in less than six years. A polyglot (English, French, German, and Spanish) and accomplished musician (cello), Westphal was described by several Sirtris board members as having "extraordinary energy" and a "rock star" reputation in the biotechnology world. "He has an unusual combination of abilities--to understand the science and its commercial potential, and explain it all clearly in an understated way that resonates with investors," said John Freund, Managing Director and cofounder of Skyline Ventures as well as a Sirtris director at the time of the case.

Westphal had a distinctive approach to building biotech companies—his own mode of operation. Westphal's major successes, Alnylam and Momenta, both went public before many market watchers believed them to be ready for an IPO. In both cases, Westphal teamed with world-renowned authorities (Alnylam with Paul Schimmel, a prominent scientist at the Scripps Institute and biologist Philip Sharp; Momenta with Robert Langer, an MIT Institute Professor and one of the world's most prolific scientist/entrepreneurs). Westphal described the elements he looked for and the approach he took in starting and building companies:

> You need fantastic science. Second, you need a great story. Third, you need great venture capital support and lots of money. I am a big believer in raising as much money up front as possible.

[5] David Stipp, "Drink wine and Live Longer," *Fortune*, February 12, 2007.

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Applying this model and exploiting an ever-growing network among the biotech industry's prominent players, Westphal had clearly developed a successful approach to launching early stage companies and then passing the CEO's baton to a different leader. In 2003, Westphal was looking for his next investment opportunity, when he encountered Sinclair's paper in *Nature*. Westphal quickly realized that this was a novel and possibly watershed discovery *if* it could be extended to humans. Westphal phoned Sinclair to discuss the prospects of commercializing his discovery.

Launching Sirtris

At the time, Sinclair had been thinking of commercializing his work for many years. In 1999, he almost joined his mentor, Guarente, in launching Elixir Pharmaceuticals, a longevity-oriented biotech company. Several years later, as Sinclair finalized the 2003 *Nature* paper, he began exploring opportunities to form a company of his own.

Sinclair described his initial meeting with Westphal, "He came in and refused to sign a nondisclosure agreement. So, I told him I wouldn't talk to him. And he said, 'David, if I walk out of this office, I'm not coming back. So I suggest you tell me as much as you can.'" I wound up telling him more than I normally would have. It soon became apparent that he's one of the smartest people I'd ever met. But it took me months to realize that he's also a nice guy."[6]

After their initial meeting, Westphal expressed an interest in starting a company with Sinclair, but could not do so until he found someone to replace him as CEO of Acceleron, one of the companies he had launched while at Polaris. Meanwhile, Sinclair continued discussions with other investors. Westphal re-entered the picture six months later, expressing a readiness to invest and pull the venture together. Sinclair and Harvard (the owner of several pieces of intellectual property that would be licensed by Sirtris) decided to move forward with Westphal as the founding CEO.

After an agonizing decision process, Westphal chose to join Sirtris as its full-time CEO. Unlike his other start-ups, his plan this time was to remain with the company, which meant that he would be leaving behind the venture capital life and a high six-figure salary. Westphal explained his decision:

> Many people thought it was too risky to leave a successful VC career. I was taking a $500,000 paycut and my wife and I had just purchased an expensive house in Brookline close to Fenway Park. At the time, David had no data that showed resveratrol activated sirtuins in mammals or could affect mammalian glucose levels or insulin, although we hoped all that would prove true. My VC friends were telling me that I was not being rational. In some ways they were right, but I was excited about Sirtris in a way I had not yet been at my other companies.

Scientific Advisory Board

Westphal set out to attract a world-class Scientific Advisory Board (see **Exhibit 2** for details on the SAB). Virtually all early-stage biomedical companies create boards of scientific advisors. Among other roles, SAB members advise the company on matters related to scientific and experimental strategies; they sometimes assist in securing access to intellectual property produced by SAB members; and they serve as portals that keep the company abreast of developments in the broader scientific community.

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[6] David Stipp, "Drink wine and Live Longer," *Fortune*, February 12, 2007.

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Sirtris's goal was to collect the brightest scientists in the field of sirtuin research, including those who would generate IP for Sirtris and be the "eyes and ears" of the company. Sinclair explained the formation of the Sirtris SAB: "Christoph said, 'Give me a list of the top 10 people in your field.' Within a week or two, we were having conference calls with all of these people. In one case, an academic was going to start a rival company, and Christoph flew out to St. Louis and convinced him to join us instead." One observer described the Sirtris SAB and board of directors in the following terms (see **Exhibit 3** for Sirtris Board of Directors):

> Since combining forces with Sinclair, Westphal has organized what is arguably the most pedigree-rich scientific advisory board in biotech, including MIT's Sharp; Robert Langer, one of medicine's most prolific inventors, also of MIT; Harvard gene-cloning pioneer Thomas Maniatis; and Thomas Salzmann, formerly executive vice president of Merck's research arm. The group now numbers 27, among them many of the world's leading experts on sirtuins. Westphal also assembled an impressive list of directors—they include Alkermes's Pops; Aldrich, the Boston hedge fund manager and biotech veteran; and Paul Schimmel, a prominent scientist at the Scripps Research Institute in La Jolla, Calif., who has co-founded half-a-dozen biotech concerns. Westphal's right arm at Sirtris is chief operating officer Garen Bohlin, formerly a senior executive at Genetics Institute, a biotech now owned by Wyeth.[7]

Growing Sirtris

During the spring and summer of 2004, Westphal and Sinclair went on a road show to local Boston-area venture capital groups. In August, they obtained a $5 million seed (Series A) round of financing from Polaris Ventures, Cardinal Partners, Skyline Ventures and Techno Venture Management ("TVM"). Sinclair described their early efforts to raise capital:

> Christoph was very good at getting us in to talk with the majority of VCs in the Boston area over a short period of time. Although a lot of people said "no" to us, Christoph set a small window of time to invest and more than a few people started getting nervous about being left out. The short timeframe built its own momentum and helped drive interest in the company.

In November 2004, Westphal and Sinclair secured another $13 million in a Series A-prime round. (See **Exhibit 4**, which details five investment rounds; including investors, dates and investment amounts. See also **Exhibit 5**, which details pre-IPO financing at three comparable biotechnology firms.) Regarding the first two funding rounds, Westphal explained their ability to raise funds before the firm had any mammalian data:

> We were very early in terms of the science. We raised $18 million without any mammalian data, something that is almost unheard of in today's world of biotechnology. Part of our success was getting people bit by the Sirtris bug. We had a long-term vision of where we could go with the biology and the anti-aging message is extremely powerful, especially when you are talking to a bunch of aging, overweight guys who are prime targets for the drugs you want to develop.

Sirtris had several decisions to make about how to focus its drug development efforts. Sinclair had theorized that sirtuins played a role in diabetes, cancer, heart disease, neurodegenerative diseases and diseases related to mitochondrial disorders. One thing was clear: there would be no effort to claim anti-aging effects from any drug the firm produced, since the FDA did not recognize aging as a

[7] David Stipp, "Live Forever?," *Fortune*, February 5, 2007.

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disease. The firm decided to develop a drug for diabetes, a large market with epidemic numbers of type II diabetes in developed and developing countries, and an orphan drug[8] for the mitochondrial disease MELAS (mitochondrial encephalopathy, lactic acidosis, and stroke-like symptoms syndrome), a rare genetically inherited disorder. (See **Exhibit 6** for details on the global diabetes market.)

Between 2004 and 2005, Sinclair began conducting experiments that sought a connection between resveratrol and sirtuins in mice. "By early 2005, David started getting data in his lab that showed resveratrol was going to extend lifespan in a mammal," Westphal said, "At Sirtris, we had evidence you could lower glucose and insulin in mice. All of a sudden, we were getting real proof that this actually could be a drug; that this could actually be a very valuable company." Sirtris began hiring its R&D team, successfully staffing leadership positions with executives who had had long stints at large pharmaceutical and biotech companies – including Millenium Pharmaceuticals, Alkermes, and GlaxoSmithKline – identifying small molecules, developing drugs and advancing drugs through clinical trials.

With their mammalian data in hand, Sinclair and Westphal went back on the road seeking additional capital to finance Sirtris's R&D efforts. In March 2005, Sirtris closed on a $27 million Series B-round of financing.

In the next year, Sirtris made three significant advances. (See **Exhibit 7** for a timeline of Sirtris scientific proofs of principle.) First, the company created a formulation for SRT501 that kept resveratrol in its active form and increased its absorption into the bloodstream. The result was that SRT501 could get five times more resveratrol into the blood stream than the best other preparations currently available. Second, Sirtris began conducting clinical trials in India, assessing SRT501 as a diabetes therapy. The transition from an R&D-only company to a clinical-stage company was, as Westphal remarked in a press release, "an important milestone in our plan to develop a rich pipeline of therapeutics to treat diseases of aging." Third, and perhaps most significantly, Sinclair completed several experiments examining the effects of high doses of resveratrol on obese mice. In one experiment, middle-age mice fed high calorie diets with resveratrol were compared to a control group fed similar diets but without resveratrol. Remarkably, the mice fed resveratrol could run further; were leaner; and, lived 30% longer than the high fat, no-resveratrol mice in the control group. (See **Exhibit 8** for a picture of mice on diets with and without the drug.)

Sirtris researchers were exhilarated by these findings in part because the data suggested that sirtuins could play a role in managing or even delaying the onset of type-II diabetes. They were also excited because in the past experimental data with the rat model (Zucker-fa/fa) had tended to foreshadow a higher probability of success for drugs in human trials.

In April 2006, Sirtris closed on a $22 million Series C round of financing, and obtained a $15 million line of venture debt from Hercules Technology Growth Capital.

Throughout 2006, Sirtris continued to gain momentum as Sinclair's research made its way into high profile academic journals, newspapers, and other media. In June 2006, Sirtris announced that it had successfully completed dosing eighty-five human subjects in a Phase 1 safety and pharmacokinetic trial of SRT501. In October 2006, an article about Sinclair's work appeared on the

[8] Orphan drug status was a special designation by the FDA that granted drug makers a seven-year marketing monopoly along with tax reductions for an approved drug for diseases afflicting less than 200,000 people. The purpose of the orphan drug classification was to provide an incentive to drug makers to focus some of their R&D on smaller, less profitable markets. Some of the more well-known targets of orphan drugs include cystic fibrosis and snake venom.

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front page of the *Wall Street Journal*. The following month, papers published in *Cell* and *Nature* demonstrated that resveratrol increased the stamina of mice two-fold and significantly extended their lifespans. The 2006 *Nature* article also demonstrated that sirtuin activation increased within the cell the number of functional mitochondria, the powerhouses that sustain a range of cellular activities including glucose metabolism.

The discovery that sirtuin activation increased the number of functional mitochondria was, Sinclair suggested, "quite intriguing" since the number of functional mitochondria was known to decline with age. Moreover, it was well-known in the scientific community that people with above average numbers of functional mitochondria, such as famed cyclist Lance Armstrong, had above average stamina levels.

In late fall of 2006, Sinclair received an unsolicited email from Red Sox owner John Henry requesting a meeting. Westphal described the meeting with Henry:

> He visits David and me here at Sirtris. And he's a very shy, wonderful gentleman. After we present him the company, he says, "How can I be helpful to you?" And I look at him, and I say, "I think you could invest $50 million in the company." And he says, "I don't think I can do $50 million, but I think I can do $20 million." And I said, "Can we close in two weeks?" John teamed up with Peter Lynch, the legendary former fund manager of Fidelity's Magellan Fund, who did extensive due diligence, and they said, "OK." Everyone wanted in after that.

By February 2007, Westphal closed on another round of financing, raising $35.9 million, from company executives, venture firms that had contributed to previous rounds, as well as John Henry and Peter Lynch. Westphal explained their fundraising success:

> We've always been able to raise money, I think because we had money. We weren't desperate. They [potential investors] knew that they couldn't get away with trying to hammer us on the valuation, because if we don't get the valuation we want we just won't raise the money.

Moving Forward

Nutraceuticals

Westphal and Sinclair had a long-running debate over the commercial opportunity represented by their proprietary formulation of resveratrol (SRT501). Sinclair had a strong belief that there would be great public demand for this formulation long before the results of the clinical studies on SRT501 were completed, and that Sirtris, one way or another, should start selling SRT501 to the public as a nutraceutical. Sinclair received an average of 30-50 emails a day from people requesting information on how to obtain resveratrol.

For a biotech company focused on drug development, a Sirtris foray into nutraceuticals would not be unprecedented. For instance, Sigma-Tau Pharmaceuticals, a 50-year old Italy-based drug company specializing in rare diseases, had developed a nutraceuticals business around its FDA-approved drugs for metabolic and renal conditions (carnitine deficiency) as well as cancer (antineoplastic therapy). Sigma-Tau had two nutraceutical divisions and sold physician-recommended and clinically tested dietary supplements for patients with ulcerative colitis or irritable bowel syndrome.

The opportunity in nutraceuticals might be substantial. The global nutraceuticals market, which included sales of health and nutritional supplements, such as vitamin C and fish oil, had estimated

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annual sales of $120 billion, and was growing at a compound annual growth rate of 7%.[9] For example, the market for glucosamine chondroitin, a joint-health supplement most commonly taken by the elderly, exceeded $1 billion, and had been expanding at a double-digit rate.[10] The economics of the nutraceuticals marketplace were compelling. Total manufacturing costs for small molecule drugs were typically low, often in the range of 25 cents per pill or less. Other costs would depend on the commercialization strategy, of which there were numerous options. Current vendors of formulations of resveratrol that were far less bioavailable than SRT501 were charging prices in the vicinity of $1 per capsule.

Westphal, however, had doubts about whether the timing was right for such a venture, and whether the firm should even market a nutraceutical under its corporate brand. There were several issues:

The nutraceuticals market was unpredictable. It was, by and large, unregulated. No FDA approval was necessary for selling supplements. No evidence was necessary to prove a product's effectiveness or even its composition. Sinclair had tested resveratrol pills and found that some brands on the market had no active resveratrol at all. Other brands had only trace amounts of active resveratrol, far too little to have any meaningful effects on humans. Even if Sirtris had a scientifically proven product, it was not clear that science alone would be enough to differentiate Sirtris's offering from the dozen or so resveratrol supplement providers. How would the company distinguish itself? A final concern was the potential for consumer allegations about resveratrol's safety. "The potential that someone could attribute a death to SRT501 consumption could easily derail a nutraceuticals business," said Sinclair.

Another issue concerned Sirtris's identity. Was the company a scientifically rigorous enterprise focused on developing FDA-approved medicines that physicians would prescribe to improve the health of patients with aging-related disorders? Would the corporate brand suffer if it started selling 501 as a nutraceutical, especially if it became, in part, a nutraceutical retailer like the makers of glucosamine? On one hand, the Sirtris office walls were plastered with pictures of Sinclair and Westphal with Nobel prize winning biologists and luminaries from the venture capital world. On the other hand, there were the pictures of Sinclair and Westphal with celebrities, including Barbara Walters and Mel Gibson. Rich Aldrich, one of the company's original investors and current board member, summarized the issues this way: "The Sirtris story is a balance. It's carefully constructed from the core science and Christoph and David's public outreach. It's not clear if a nutraceutical approach will taint that story or extend it."

Even if they did consider the nutraceuticals business, what was the business case for market entry? How much of a return would warrant their participation in the market? And how would market entry be achieved? Should the company set up a subsidiary or create a spin-off company devoted to SRT501? Would the subsidiary have the Sirtris name, e.g., Sirtris Health, or would the subsidiary have a different name, as Lexus is to Toyota? Some Sirtris executives were favoring a wholly owned nutraceutical subsidiary with a different name, in order to make clear that the Sirtris brand was focused on drugs based on their NCE development platform. "Our long-term investors like this option because they are in this company for the NCEs. They're not in it for the 501 data," said Westphal.

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[9] http://biz.yahoo.com/bw/080220/20080220005585.html?.v=1. Accessed February 22, 2008.

[10] Eric Nagourney, "Study Sees Little Benefit in Chondroitin for Arthritis," *New York Times*, April 17, 2007.

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Yet another issue was retail format. How would a nutraceutical be sold: its own retail stores, the Internet, supplement stores, such as GNC? Would developing its own retail distribution take away from its drug development? Would partnering with a GNC retailer reduce control over the brand?

Pharmaceutical Deal

Throughout 2006, new evidence emerged from laboratories around the world confirming Sinclair's hypothesis of a connection between resveratrol, sirtuins and metabolic activity in mammals. Several pharmaceutical companies began talking with Westphal about a possible drug development partnership. He anticipated that the terms of a deal could include a significant upfront cash payment, an equity purchase agreement, non-milestone-based (i.e., guaranteed) R&D support that would be likely to step up over a four to five year period, and payments tied to clinical development milestones. Sirtris would also receive royalties on sales of any drugs resulting from its SIRT activation program (See **Exhibit 9** for more details on the terms of a deal that might be possible and **Exhibit 10** for details of three deals between big pharma and other private biotechnology firms.) As an alternative, the team also believed that they could ink a deal with terms similar to those in **Exhibit 9** but in which the pharmaceutical partner would take a 51% equity stake in the company.

As Westphal, Sinclair, and Dipp resumed the conversation they had been having in early 2007 about the reasons for and against a deal, one positive aspect that continued to come up was that having a drug development deal was often a condition of having a successful initial public offering. Sirtris executives were hoping to take the company public at some point in the not-too-distant future. A successful IPO could deliver some of the financial resources a company needs to move drug development from the laboratory through clinical trials. "The typical biotech playbook says to get a partnership deal done, then file for an IPO. Public investors are often reassured by the prior involvement of a pharmaceutical company," said Westphal. Several board members intent on satisfying the company's capital needs had already voiced their interest in exploring whether a deal could be completed on attractive terms. Jeff Leiden, a life sciences venture capitalist and the former president and chief operating officer of Abbott Pharmaceuticals and a friend of Westphal's, noted that a typical biotechnology company would not be a strong IPO candidate until it had developed some clinical data, successfully made headway on two different research programs, established intellectual protection around its discovery of one or more new chemical entities, and signed at least one significant deal with a pharmaceutical company.

Another reason to contemplate a deal was that it might be a relatively inexpensive source of additional capital. Pharmaceutical partners were often willing to purchase an equity stake at a premium to VC investors. As well, they might be willing to finance Sirtris's R&D programs.

There were reasons not to do a deal. Westphal and Sinclair were building a company that they hoped would have great medical and investment impact. Even though Westphal had been the founding CEO of several companies, this was the first company that he had actively managed beyond the first two years. This was also Sinclair's first company. How would a deal affect the founders' control over the company and its future?

Another reason to consider deferring a deal was Sirtris's promising development of new chemical entities (NCEs). Resveratrol is a naturally occurring substance and because it was already on the market as a nutraceutical, intellectual property protection related to it would be limited to "methods of use" patents, which would cover the use of resveratrol to treat particular diseases, such as diabetes, as well as formulation work. Sirtris had recently made strides in synthesizing new compounds that could be as much as 1000 times more potent than resveratrol at activating SIRT1 and that would be eligible for NCE composition of matter patents. Partnering with a large pharmaceutical company would require out-licensing these new compounds without knowing their full value. If

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Sirtris waited for new NCE data to arrive, it might be able to arrange a more lucrative deal than what might have been doable in early 2007. Of course, if the NCE data came in and it did not produce the results Sirtris was expecting, the pharma deal terms would be substantially worse, assuming that the pharmaceutical companies remained interested at that point.

In-Licensing or Expanding the Scientific Base

From the very beginning of the company, Sirtris executives had had an internal debate over how much of the company's resources to focus on SIRT1 versus alternative sirtuin targets. The biotechnology industry was littered with companies and drug development projects that had stalled in moving from mouse studies and toxicity screens to human trials. Several board members were advocating that the firm diversify its product development platform. There were two main alternatives. One was to investigate the six other human sirtuins. The other was to in-license from another biotechnology firm a compound that had a better known mechanism of operation.

The study of sirtuins was still in its infancy, so investigating the clinical possibilities of other sirtuins would require a great deal of basic research, financing and time. (See **Exhibit 11** for Sirtris financial data, 2004-2006.) Even so, the clinical role of the other human sirtuins offered tantalizing commercial options. SIRT3, for instance, was found in mitochondria and was considered a potential drug target, but little was currently known about its functional role. Developing a research platform based on SIRT3 might complement the company's own drug development efforts in other mitochondrial disorders, including diabetes and MELAS.

Several members of the SAB and the board of directors considered the in-licensing option to be an appealing alternative, although others disagreed. It would balance their investment in SIRT1, which was absorbing 90% of the firm's R&D expenditures. Dr. Dipp explained that, after investigating more than one hundred potential compounds to in-license, Sirtris had found a few anti-diabetic compounds that had better characterized effects than sirtuin-activating compounds. "It would be what we call "me-too" drugs. We know how they work, and if we could get them on the market they would get at least a small percentage of market share. It's something to have in your back pocket."

When the firm launched in 2004, Sinclair and Westphal debated whether to in-license a compound and decided against doing so. "I was the only person at the company," said Sinclair, "who thought that SIRT1 activation was the right bet to make. I told Christoph, 'don't stop it until you know it's wrong.' If I'm wrong, find out sooner than later, and then in-license something." Westphal offered another view, "For the first eight months of this company, I was sitting there like a venture guy thinking that resveratrol will not be a great drug. It's a great story, but we'll have to bring in other stuff to build the company around."

Even though new experimental data seemed to confirm that resveratrol activated SIRT1, and that SIRT1 activation could be clinically important, Sinclair, Westphal, Bohlin and Dipp continued to debate whether to focus the firm's time, money, and other resources on that one target or divert more of the firm's resources to additional targets, including non-sirtuins. They still did not have evidence that SIRT1 had the effects in humans that Sinclair believed they would one day see.

Conclusion

After discussing these three issues for several hours, Sinclair and Westphal decided to summarize their views on the decisions they needed to make.

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At a general level, they remained convinced that sirtuin-activating drugs, if they could be successfully developed, would have a revolutionary impact on human disease. However, they recognized that Sirtris was still many years from completing development of these drugs, much less manufacturing and selling them. To reach that distant point would require successfully navigating technical and regulatory hurdles that had stymied the majority of other biotechnology companies at similar points in their history. According to a pharmaceutical industry association report, only one in five compounds entering clinical trials gained FDA approval.[11] And, only 30% of approved drugs recovered the average development cost of a new medicine.[12] Given all of the unknowns about what could happen Sinclair and Westphal described several options for addressing the risks they faced:

One approach would be to fully "hedge their bets:" Sirtris could try to complete a pharmaceutical deal, in-license a compound with better known effects, and consider a nutraceuticals business around SRT501.

Another approach would be to "swing for the fences" (or, in a frequently used metaphor in the industry, "one shot at gold"). This would continue the firm's focus on a sirtuin-activating drug development platform. If successful, Sirtris could become, as one pharmaceutical executive suggested, "the Google of biotech." However, an IPO would be less likely in the interim, since markets often prefer that biotech firms have a validating deal with a large pharmaceutical company.

A third approach would be a "middle of the road" path that incorporated some hedging, such as pursuing an in-licensing deal, but also accepted some risk, e.g., deferring a potential pharmaceutical deal. Alternatively, Sirtris could try to complete a pharmaceutical deal now, but forego in-licensing and the nutraceuticals project.

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The company seemed to have arrived at a critical juncture in its development. What approach should Sinclair and Westphal take? And why?

[11] PhRMA Pharmaceutical Profile Industry Report, 2007. http://www.phrma.org/files/Profile%202007.pdf, Accessed February 28, 2008.

[12] PhRMA Pharmaceutical Profile Industry Report, 2007. http://www.phrma.org/files/Profile%202007.pdf, Accessed February 28, 2008.

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Exhibit 1 Anti-Aging Science Timeline

1884: William Jones MD reports that a fasting spider lives unusually long, 204 days. [Longevity in a fasting spider, *Science* 3: 4, Jan. 4, 1884]

1934 Clive McCay and Mary Crowell of Cornell University report delayed aging in rats on limited calorie diets. [McCay, C. M., and M. F. Crowell. 1934. Prolonging the life span. *Science Monthly* 39:405–414]

1986 Roy Walford and Richard Weindruch show limited-calorie diet produces *"youthful"* old mice. [Walford R. Weindruch R, *Journal Nutrition* 116: 641, 1986; The Retardation of Aging and Disease by Dietary Restriction, 1988]

2000: Leonard Guarente of MIT mimics calorie restriction's prolonged lifespan effects in yeast cells. [*Science* 289: 2126, 2000]

2003 David Sinclair of Harvard extends life of yeast cells with resveratrol. [*Nature* 425: 191, Sept. 11, 2003]

2004 Marc Tatar of Brown University and David Sinclair of Harvard report resveratrol extends life of roundworms. [*Nature* 430: 686, July 14, 2004]

2006: Italian researchers are first to report resveratrol prolongs the lifespan of a vertebral animal (cold-water fish). [*Current Biology* 16: 296, Feb. 7, 2006]

2006 David Sinclair and colleagues demonstrate resveratrol extends life of warm-blood mammal (mice) and overcomes deleterious effects of a high-fat diet. [*Nature* 444: 337, Nov. 1, 2006]

2006 *JAMA*. 2006 Apr 5;295(13):1539-48.

2006 Lagouge, M. et al., Resveratrol improves mitochondrial function and protects against metabolic disease by activating SIRT1 and PGC-1alpha. Cell. 2006 Dec 15;127(6):1109-22.

2007 Civitarese, A.E. et al., Calorie Restriction Increases Muscle Mitochondrial Biogenesis in Healthy Humans. *PLoS Med.* 2007 Mar 6;4(3).

Source: Adapted from http://www.longevinex.com/article.asp?story=Time%20Of%20Research%20Involving%20Calorie%20Restriction%20and%20Longevity, accessed February 25, 2008; and Sirtris documents.

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Exhibit 2 Sirtris Scientific Advisory Board

BioPharma Expertise	Sirtuin Expertise
Tom Salzmann, Co-Chair SAB Former EVP Merck	Biochemistry John Denu, Wisconsin Anthony Sauve, Cornell Vern Schramm, AECOM David Sinclair, Co-Chair SAB, HMS Eric Verdin, UCSF
Julian Adams President and CSO, Infinity	
Peter Hutt Former FDA General Counsel	Mouse Genetics, Phenotypes Fred Alt, HMS, NAS, HHMI Johan Auwerx, ICB
Bob Langer MIT, NAS, NAE, IOM, Co-Founder Momenta, AIR	Structural Biology Cynthia Wolberger, JHU, HHMI
Tom Maniatis Harvard, NAS, Co-Founder GI, Acceleron	Links to Disease Ron Kahn, Joslin, NAS Jeff Milbrandt, Washington University Jerrold Olefsky, UCSD Pere Pulgserver, HMS Eric Ravussin, Pennington Institute Li-Huei Tsai, MIT, HHMI
Phil Sharp Co-Founder Biogen and Alnylam, NAS, Nobel Prize	
Ted Sybertz SVP Genzyme, Schering (Zetia)	
Eric Gordon Former head of Medicinal Chemistry at Bristol-Myers, Founder of Vicuron Pharmaceuticals	
Chris Walsh Harvard, NAS, IOM, Genzyme, Vicuron	

Source: Company.

HHMI = Howard Hughes Medical Institute
ICB = Mouse Clinical Institute in Strasburg, France
NAS = National Academy of Sciences
NAE = National Academy of Engineers
AECOM = Albert Einstein College of Medicine

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Exhibit 3 Sirtris Board of Directors (as of February 2007)

Name	Background
Richard Aldrich (F)	Founder, RA Capital
John Clarke	Managing General Partner, Cardinal Partners
Alan Crane	Momenta
John Freund	Managing Director, Skyline Ventures
Stephen Hoffman, Ph.D., M.D.	Managing Director, TVM Capital
Wilf Jaeger	Three Arch Partners
Stephen Kraus	Bessemer Venture Partners
Richard Pops (F)	Chairman, Alkermes
Paul Schimmel, Ph.D. (F)	Professor, The Scripps Research Institute
David Sinclair, Ph.D. (F)	Associate Professor, Harvard Medical School
Christoph Westphal, M.D., Ph.D. (F)	CEO and Vice Chair, Sirtris Pharmaceuticals, Inc.

Source: Company.

F = Founder

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Exhibit 4 Sirtris Investors and Investment Rounds

Round	Date	Investors	Investment	Pre-$ Valuation
A	Aug. 2004	Polaris Venture Partners, Cardinal Partners, Skyline Ventures and TVM	$5 M	$2.8 M
A1	Oct. 2004	Polaris Venture Partners, Cardinal Partners, Red Abbey, Skyline Ventures, TVM and The Wellcome Trust	$13M	$10.9M
B	Mar 2005	Three Arch Partners, Cargill Ventures, Novartis Bioventures Fund, Polaris Venture Partners, TVM, Red Abbey, Cardinal Partners, Skyline Ventures, and The Wellcome Trust	$27 M	$32.5M
C	Apr 2006	Bessemer Venture Partners, Genzyme Ventures, QVT Fund LP, Alexandria Real Estate Equities, Inc. Polaris Venture Partners, TVM Capital, Cardinal Partners, Skyline Ventures, Three Arch Partners, The Wellcome Trust, Novartis Bioventures Fund, Cargill Ventures, Cycad Group, Hunt Ventures, and Red Abbey	$22 M ($22 million in Series C-equity placement, plus $15 million in venture debt financing from *Hercules Technology Growth Capital)*	$95.4M
C1	Feb. 2007	John Henry Trust, Peter Lynch and several investors from previous rounds	$35.9 million	$184M

Source: Company.

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Exhibit 5 Pre-IPO Financing of Comparable Biotechnology Companies

Company	Round/Date	Investors	Investment
Anesiva (IPO 2004)	2Q2001	InterWest Partners, Alta Partners, J.P. Morgan Partners	$13 million
	2Q/2002	Bear Stearns Health Innoventures, Alta Partners, HBM BioVentures, J.P. Morgan Partners, InterWest Partners, MIC Capital, China Development Industrial Bank (CDIB Ventures)	$50 million
	4Q/2003	Bristol Myers Squibb	$45 million
Cytokinetics (IPO 2004)	2Q/1998	Mayfield, Sevin Rosen Funds, Individual Investors	$5.3 million
	3Q/1999	International BM Biomedicine Holdings, Vulcan Capital, Mayfield, Sevin Rosen Funds, NMT New Medical Technologies, Duke University, Individual Investors	$20 million
	4Q/2000	Credit Suisse, Alta Partners, Lombard Odier Darier Hentsch, Mayfield, Sevin Rosen Funds, Vulcan Capital, NMT New Medical Technologies, Duke University, Individual Investors	$50 million
	2Q/2001	GlaxoSmithKline	$14 million
Momenta	2Q/2002	Cardinal Partners, Polaris Venture Partners	$4.4 million
(IPO 2004)	2Q2003	Atlas Venture, MVM Limited, Polaris Venture Partners, Cardinal Partners	$19 million
	1Q2004	Mithra Group, Polaris Venture Partners, Atlas Venture, MVM Limited, Cardinal Partners	$20 million

Source: Adapted from Growthink Research.

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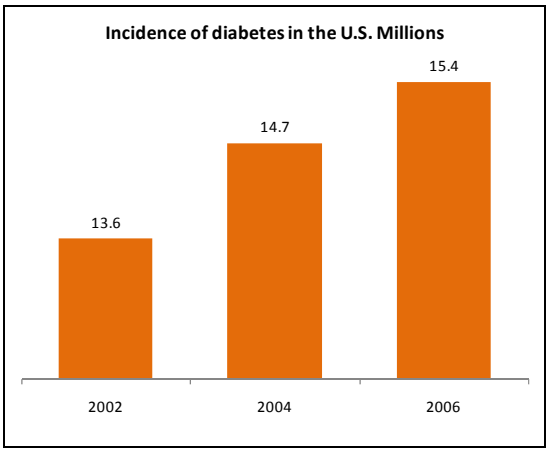
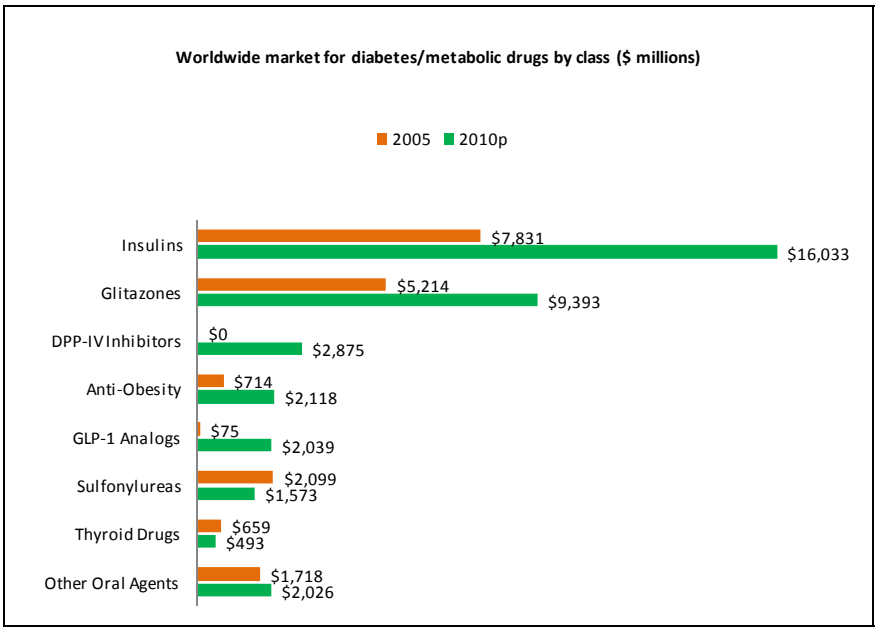
Exhibit 6 Diabetes Market Data





- 90% of the total diabetic population has Type 2 Diabetes
- Total direct and indirect costs associated with diabetes were over $100 billion in 2002

Source: Company.

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Exhibit 7 Proof of Principle for SIRT1 Activators

Pre-clinical Therapeutic	*Nature* 2006 Nov 16:444(7117):337-42. *Cell*. 2006 Dec. 15; 127(6):1109-22. Unpublished
Human Physiological	*PLoS Medicine*. 2007 Mar 6:4(3). *JAMA*. 2006 Apr 5;295(13) 1539-48.
Human Genetic	*Cell*. 2006 Dec 15;127(6):1109-22.

Source: Company.

Exhibit 8 Effects of High Doses of Resveratrol in Mice

Two mice fed the same high-calorie diet in a Sirtris-sponsored study: The svelte one on the left received high doses of resveratrol.

Source: Company.

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Exhibit 9 Potential Sirtris Pharma Deal Terms

- Five year term.
- 19.9% OR 51% equity stake at a 50% premium to most recent share price.[a]
- $75 million upfront for an exclusive option to join Sirtris in developing and marketing compounds from its SIRT1 Activator Program.
- 5 years of guaranteed R&D support totaling $100 million. Payments to step up over time. $10 million in year one.
- A combination of royalties, possibly manufacturing profits, and co-promotion fees that equate to approximately a 50/50 split of profits in the U.S. This is a significant point for Sirtris since the SIRT 1 activator program is a core program, and the one that represents 90% plus of the firm's value.
- The pharmaceutical company will lead marketing and country specific development ex-U.S., Sirtris to receive substantial, double digit royalties on ex-U.S. sales.
- Roughly $200 million in milestones concurrent with risk reducing progress. Roughly 15% upon successful completion of safety/PK of a SIRT 1 activator NCE in humans; Roughly 25% based on observation of glucose effects in phase 1b of NCE in man; Roughly 30% upon successful completion of a phase 2a efficacy study for an NCE in man; and roughly 30% upon completion of phase 2b studies.

a: Two different equity stakes were under discussion—19.9% or 51% of Sirtris's equity. All other terms would remain the same in both scenarios.

Source: Company.

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Exhibit 10 Comparison Agreements between Comparable Biotechnology Start-ups and Large Pharmaceutical Companies

Company	Deal Partner	Agreement	Terms
Anesiva is a late-stage biopharmaceutical company that seeks to be the leader in the development and commercialization of novel therapeutic treatments for pain. The Company has four drug candidates in clinical development for multiple potential indications. IPO in 2004 (formerly called Corgentech—this was the name when it went public).	Bristol-Myers Squibb Company	Jointly develop and commercialize Corgentech's E2F Decoy (edifoligide sodium), a first-of-its-kind E2F Decoy treatment currently in Phase III development for the prevention of vein graft failure following coronary artery bypass graft (CABG) and peripheral artery (i.e., leg) bypass graft surgery.	Bristol-Myers Squibb will make an initial payment to Corgentech of $45 million comprising cash and an equity investment in Corgentech, with the potential for an additional $205 million in clinical and regulatory milestone payments. Bristol-Myers Squibb and Corgentech will share development costs in the U.S. and Europe based on a pre-agreed percentage allocation.
Cytokinetics is a biopharmaceutical company dedicated to the discovery, development and commercialization of novel classes of small-molecule therapeutics, particularly in the field of cytoskeletal pharmacology. IPO in April 2004	GlaxoSmithKline	A broad strategic collaboration to discover, develop and commercialize novel small-molecule therapeutics targeting mitotic kinesins for applications in the treatment of cancer and other diseases.	GSK has committed funding of approximately $50 million over the minimum 5-year research term, including a $14 million upfront cash payment and a $14 million purchase of Cytokinetics preferred stock. In addition, GSK could make milestone payments to Cytokinetics ranging from $30-50 million per target for products directed to each of over 10 mitotic kinesins that will be the subject of collaborative activities.
Momenta Pharmaceuticals was founded based on proprietary technology developed at MIT that enables high throughput, detailed characterization and engineering of sugars. IPO in June 2004	Sandoz/Novartis	A strategic alliance covering joint product development and commercialization in the area of complex pharmaceutical products. The collaboration will apply Momenta's novel technological capabilities related to complex sugars and the leadership of Sandoz in the generic pharmaceuticals industry to pursue the joint goal of commercializing products. Under the terms of the agreement, Sandoz and Momenta will jointly manage product development and commercialization.	Sandoz has committed $600,000 in an upfront cash payment and has the right to purchase $5-10 million in Momenta equity. R&D payments estimated to be $12 million and up to $50 million in contingent payments to accompany development milestones.

Source: Adapted from Recombinant Capital.

NOTE: All deals inked when biotech was still private.

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Exhibit 11 Sirtris Financial data 2004-2006 (in thousands, except share and per share amounts)

	Period from March 28, 2004 (date of inception) through December 31, 2004	Year Ended December 31, 2005	Year Ended December 31, 2006	Period from March 25, 2004 (date of inception) through December 31, 2006
Statement of operations data:				
Revenue	$ —	$ 68	$ —	$ 68
Operating expenses:				
Research and development	1,190	7,062	14,242	22,494
General and administrative	699	3,865	4,340	8,904
Total operating expenses	1,889	10,927	18,582	31,398
Loss from operations	(1,889)	(10,859)	(18,582)	(31,330)
Interest income	45	1,143	2,447	3,635
Interest expense	—	—	(878	(878
Net loss	$(1,844)	$(9,716)	$(17,013)	$(28,573)
Net loss per share—basic and diluted	$ (0.93)	$ (3.16)	$ (3.52)	
Weighted average number of common shares used in net loss per share—basic and diluted	1,995,468	3,087,716	4,854,646	
Pro forma net loss per share—basic and diluted			$ (0.20)	
Shares used in computing pro forma net loss per share—basic and diluted			85,603,228	

Source: Company.

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Appendix Wall Street Journal Cover

THE WALL STREET JOURNAL.

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Youthful Pursuit

Researchers Seek Key to Antiaging In Calorie Cutback

A Controversial Hypothesis Draws Scientists, Investors; Will It Work in Humans?

Fighting Fat in Lab Mice

By David Stipp

In the 1930s, researchers stumbled onto a surprisingly simple way to slow the biological forces of aging: cutting normal calorie intake by about a third. Scientists found it boosts animals' life spans by 30% to 40%, and considerable evidence suggests that calorie restriction, or CR, would slow human aging too.

David Sinclair

But only steely ascetics could hack its hunger pangs. So the finding remained a little-known curiosity in the back halls of science.

Now a coterie of scientists and biotech ventures are rekindling interest in CR as they try to mimic its antiaging effects with medicines. It is still a highly speculative quest, and many researchers fret that it hasn't completely shaken its association with centuries of dubious nostrums to slow aging, from inhaling virgins' breath to eating gold to implanting monkey glands.

Much of the new focus is on a substance in red wine called resveratrol. The interest in it started three years ago when a group led by Harvard Medical School biologist David Sinclair reported that it boosted yeast cells' life span by 70% via a mechanism resembling CR. He later co-authored a study showing that it also boosts life span in fruit flies and roundworms. But his tendency to make bold leaps based on tentative data has also sparked intense controversy. One big question: Does he really understand the workings of CR well enough to mimic them in a drug?

Last spring, Italian scientists reported that resveratrol boosted life span more than 50% in a kind of short-lived fish. Intriguingly, fish on resveratrol had much faster swimming speeds as they aged, and spent far more time moving around, than did undosed control fish.

At least two groups of researchers are now testing whether resveratrol can extend life span in mice—the first such studies in mammals. At a meeting of the American Aging Association in June, Dr. Sinclair and colleagues presented preliminary results from a study showing that resveratrol had "CR-like protective effects" against the buildup of fatty deposits in the livers of mice on high-calorie diets. That suggests that resveratrol could lead to new drugs for diseases of aging associated with rich diets, such as adult-onset diabetes.

Leonard Guarente

A company that Dr. Sinclair co-founded in 2004, Sirtris Pharmaceuticals Inc., of Cambridge, Mass., has begun testing a resveratrol-based drug in diabetic patients. It has raised $82 million from venture capitalists, a hefty sum for an early-stage biotech. (Sirtris's chief executive, Christoph Westphal, is married to a reporter for this newspaper.)

It faces competition from Elixir Pharmaceuticals Inc., also based in Cambridge, which Dr. Sinclair's former mentor, Massachusetts Institute of Technology biologist Leonard Guarente, co-founded in 1999 to develop drugs based on gene variants that slow aging. The niche also includes BioMarker Pharmaceuticals of Campbell, Calif., and LifeGen Technologies of Madison, Wis., both of which focus on mimicking CR with drugs.

The companies hope to develop therapies for diseases, not antiaging pills. One reason is that the Food and Drug Administration doesn't recognize aging as a problem warranting treatment. But if a drug could retard aging, it might delay the onset and possibly the progression of age-related diseases. "When you slow aging," says University of Illinois epidemiologist S. Jay Olshansky, "you push a host of diseases to later ages at one fell swoop—cancer, heart disease, Alzheimer's, diabetes, as well as everything else that's negative about growing older."

Some researchers believe antiaging drugs could also improve health in late life—rather than prolong misery—letting people stay in relatively good shape until a swift demise. Their case rests partly on the svelte, energetic look of old animals on CR. "Often it's hard to identify the cause of death" in post-mortem studies on such animals, says Richard Weindruch, a University of Wisconsin CR researcher. "The only apparent problem is that they died."

Still, some experts on aging doubt that enough is known about CR to guide the development of drugs that mimic its effects. "We know a lot about CR's effects," says Edward Masoro, a leading gerontologist. "But what bothers me is that I don't think we've figured out CR's basic mechanism yet."

Dr. Sinclair's idea that resveratrol mimics CR has come under heavy fire. His main adversaries are two researchers who used to rub elbows with him when they all studied together with MIT's Dr. Guarente. The skeptics maintain that resveratrol's mode of action is still murky; instead, they are looking at other mechanisms that may account for how CR works.

The resveratrol doses used in the life-span-extension studies in animals were far higher than the amount people can get by drinking wine—they were roughly equivalent to hundreds of glasses a day. Resveratrol is available as a dietary supplement, but to replicate the doses used in the studies, a person would need to take scores of pills a day. (Sirtris says it is developing prescription drugs that work like resveratrol but are hundreds of times more potent.) The dietary supplements haven't been tested in clinical trials, so their efficacy isn't proven, nor is it clear what dose might make people live healthier or longer. And although they seem safe at modest doses, megadoses may not be.

Nevertheless Dr. Sinclair, a 37-year-old Australia native, thinks taking small doses over time may yield health benefits and has been taking the supplements for three years.

The story of resveratrol has its roots in scientists' increased understanding of CR. In 1989 researchers theorized that it acti-

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Source: David Stipp, "Researchers seek key to antiaging in calorie cutback," *Wall Street Journal*, October 30, 2006.

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